

EXIST 2026

sEXism Identification in Social neTworks

Lab Guidelines

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Source: unsplash

<https://nlp.uned.es/exist2026/>

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Task Description

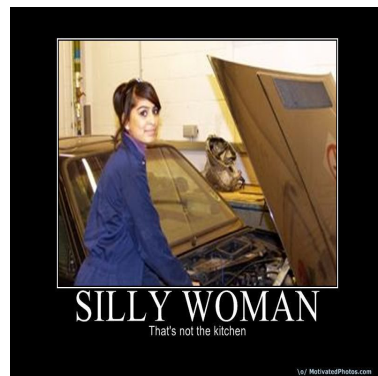
Building upon the **EXIST 2025 dataset**, this edition focuses exclusively on **multimedia formats**, comprising six experimental subtasks applied to **images (memes)** and **videos (TikToks)**. Participants are challenged to address three main objectives: **sexism identification (x.1)**, **source intention detection (x.2)**, and **sexism categorization (x.3)**.

A groundbreaking feature of this lab is the integration of **Human-Centered AI principles**. In the new experimental framework introduced in **EXIST 2026**, selected subjects were exposed to the multimedia content while their physiological and behavioral responses were continuously recorded. These **multimodal signals** (including **eye tracking**, **heart rate**, and **EEG**) enrich the traditional annotation labels, providing a deeper window into how users unconsciously process and react to sexist content in **English and Spanish**.

Although the task numbering may seem unusual, it has been kept this way to ensure consistency with the structure of the 2025 dataset and the organization of the EvALL repository.

SUBTASK 2.1: Sexism Identification in Memes

This is a binary classification subtask consisting of deciding whether or not a given meme is sexist, contains sexist expressions or behaviours (i.e., it is sexist itself, describes a sexist situation or criticizes a sexist behaviour), and classify it according to two categories: **YES** and **NO**. The following figures are some examples:



(a) YES



(b) NO

SUBTASK 2.2: Source Intention in Memes

Once a message has been classified as sexist, the second subtask aims to categorize the meme according to the intention of the author, which provides insights in the role played by social networks on the emission and dissemination of sexist messages. Due to the characteristics of the memes, systems should only classify memes with **DIRECT** or **JUDGEMENTAL** labels.

- **DIRECT**: the intention was to write a message that is sexist by itself or incites to be sexist.

- **JUDGEMENTAL:** the intention was to judge, since the tweet describes sexist situations or behaviours with the aim of condemning them.

The following figures are some examples of them, respectively.



(a) Direct



(b) Judgemental

SUBTASK 2.3: Sexism Categorization in Memes

Many facets of a woman's life may be the focus of sexist attitudes including domestic and parenting roles, career opportunities, sexual image, and life expectations, to name a few. Automatically detecting which of these facets of women are being more frequently attacked in social networks will facilitate the development of policies to fight against sexism. According to this, each sexist meme must be categorized in one or more of the following categories

- **IDEOLOGICAL AND INEQUALITY:** The text discredits the feminist movement, rejects inequality between men and women, or presents men as victims of gender-based oppression.
- **STEREOTYPING AND DOMINANCE:** The text expresses false ideas about women that suggest they are more suitable to fulfill certain roles (mother, wife, family caregiver, faithful, tender, loving, submissive, etc.), or inappropriate for certain tasks (driving, hardwork, etc), or claims that men are somehow superior to women.
- **OBJECTIFICATION:** The text presents women as objects apart from their dignity and personal aspects, or assumes or describes certain physical qualities that women must have in order to fulfill traditional gender roles (compliance with beauty standards, hypersexualization of female attributes, women's bodies at the disposal of men, etc.).
- **SEXUAL VIOLENCE:** Sexual suggestions, requests for sexual favors or harassment of a sexual nature (rape or sexual assault) are made.
- **MISOGYNY AND NON-SEXUAL VIOLENCE:** The text expresses hatred and violence towards women.

The following figures are some examples of categorized memes.



(a) Stereotyping



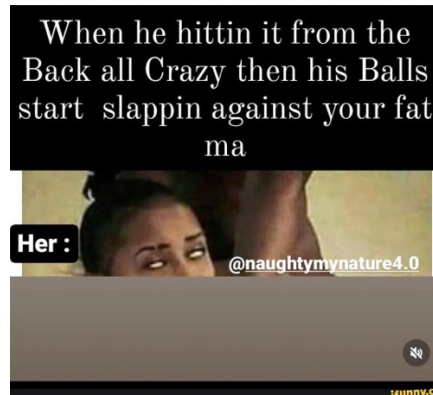
(b) Ideological



(c) Objectification



(d) Misogyny



(e) Sexual violence

SUBTASK 3.1: Sexism Identification in Videos

This subtask is the same subtask as subtask 2.1. Some examples of videos classified as YES or NO can be accessed on the website: <https://nlp.uned.es/exist2026/>

SUBTASK 3.2: Source Intention in Videos

This subtask replicates subtask 2.2 for memes, but it takes as source videos. Some examples of videos classified as DIRECT or JUDGEMENTAL can be accessed on the website: <https://nlp.uned.es/exist2026/>

SUBTASK 3.3: Sexism Categorization in Videos

This subtask aims to classify sexist videos according to the categorization provided for Task 1.3: (i) IDEOLOGICAL AND INEQUALITY, (ii) STEREOTYPING AND DOMINANCE, (iii) OBJECTIFICATION, (iv) SEXUAL VIOLENCE and (v) MISOGYNY AND NON-SEXUAL

VIOLENCE. Some examples of videos classified can be accessed on the website: <https://nlp.uned.es/exist2026/>

More details can be found at the EXIST 2026 website (<https://nlp.uned.es/exist2026/>).

Datasets Description

The EXIST 2026 Memes Dataset will be used in Task 2.X (subtasks 1–3), while the EXIST 2026 Videos Dataset will be used in Task 3.X (subtasks 1–3).

This edition builds upon the EXIST 2025 dataset and extends it with additional sensor-based information collected under a Human-Centered AI framework, enriching the original annotations with complementary physiological data.

EXIST 2026 Memes Dataset

The EXIST 2026 Memes Dataset contains more than 5,000 labeled memes, both in English and Spanish. In particular, the training set contains 3,984 memes and the test set contains 1,053 memes. Distribution between both languages has been balanced. Additionally, a small number of memes from the 2025 edition have been removed after detecting duplicated instances that were not identified during the initial dataset creation process.

The data sets are provided in **JSON format**. Each meme is represented as a JSON object with the following attributes:

1. **"id_EXIST"**: a unique identifier for the meme.
2. **"lang"**: the languages of the meme ("en" or "es").
3. **"text"**: the text automatically extracted from the meme.
4. **"meme"**: the name of the file that contains the meme.
5. **"path_memes"**: the path to the file that contains the meme.
6. **"number_annotators"**: the number of persons that have annotated the meme.
7. **"annotators"**: a unique identifier for each of the annotators.
8. **"gender_annotators"**: the gender of the different annotators. Possible values are: "F" and "M", for female and male respectively.
9. **"age_annotators"**: the age group of the different annotators. Possible values are: 18-22, 23-45, and 46+.
10. **"ethnicity_annotators"**: the self-reported ethnicity of the different annotators. Possible values are: "Black or African America", "Hispano or Latino", "White or Caucasian", "Multiracial", "Asian", "Asian Indian" and "Middle Eastern".
11. **"study_level_annotators"**: the self-reported level of study achieved by the different annotators. Possible values are: "Less than high school diploma", "High school degree or equivalent", "Bachelor's degree", "Master's degree" and "Doctorate".
12. **"country_annotators"**: the self-reported country where the different annotators live in.

13. **"labels_task2_1"**: a set of labels (one for each of the annotators) that indicate if the meme contains sexist expressions or refers to sexist behaviours or not. Possible values are: "YES" and "NO".
14. **"labels_task2_2"**: a set of labels (one for each of the annotators) recording the intention of the person who created the meme. Possible labels are: "DIRECT", "JUDGEMENTAL", "-", and "UNKNOWN".
15. **"labels_task2_3"**: a set of arrays of labels (one array for each of the annotators) indicating the type or types of sexism that are found in the meme. Possible labels are: "IDEOLOGICAL-INEQUALITY", "STEREOTYPING-DOMINANCE", "OBJECTIFICATION", "SEXUAL-VIOLENCE", "MISOGYNY-NON-SEXUAL-VIOLENCE", "-", and "UNKNOWN".
16. **"split"**: subset within the dataset the meme belongs to ("TRAIN-MEME", "TEST-MEME" + "EN"/"ES").

EXIST 2026 Videos Dataset

The EXIST 2026 Videos Dataset contains more than 3,000 labeled Tiktok videos, both in English and Spanish. In particular, the training set contains 2,524 videos and the test set contains 674 videos. Distribution between both languages has been balanced.

The data sets are provided in **JSON format**. Each video is represented as a JSON object with the following attributes:

1. **"id_Tiktok"**: a unique identifier in Tiktok platform for the video.
2. **"id_EXIST"**: a unique identifier for the video in EXIST evaluation.
3. **"lang"**: the languages of the video ("en" or "es").
4. **"text"**: the text automatically extracted from the video.
5. **"video"**: the name of the file that contains the video.
6. **"path_video"**: the path to the file that contains the video.
7. **"url"**: the url to the Tiktok video.
8. **"annotators"**: a unique identifier for each of the annotators.
9. **"number_annotators"**: the number of persons that have annotated the video.
10. **"gender_annotators"**: the gender of the different annotators. Possible values are: "F" and "M", for female and male respectively.
11. **"labels_task3_1"**: a set of labels (one for each of the annotators) that indicate if the video contains sexist expressions or refers to sexist behaviours or not. Possible values are: "YES", "NO" and "UNKNOWN".
12. **"labels_task3_2"**: a set of labels (one for each of the annotators) recording the intention of the person who created the video. Possible labels are: "DIRECT", "JUDGEMENTAL", "-", and "UNKNOWN".
13. **"labels_task3_3"**: a set of arrays of labels (one array for each of the annotators) indicating the type or types of sexism that are found in the video. Possible labels are: "IDEOLOGICAL-

INEQUALITY”, “STEREOTYPING-DOMINANCE”, “OBJECTIFICATION”, “SEXUAL-VIOLENCE”, “MISOGYNY-NON-SEXUAL-VIOLENCE”, “-”, and “UNKNOWN”.

14. “**split**”: subset within the dataset the meme belongs to (“TRAIN-VIDEO”, “TEST-VIDEO” + “EN”/“ES”).

IMPORTANT: Since labels for subtasks 2.2, 2.3, 3.2 and 3.3 are only assigned if the meme/video has been labeled as YES, the label “-” is assigned to not sexist instances in these subtasks. The label “UNKNOWN” is assigned to memes/videos for which the annotators did not provide a label. **Note that “UNKNOWN” is not a target class for classification**, and therefore should not be predicted by the systems (this labels are discarded in the evaluation phase).

For the test sets, labels for the subtasks are not provided.

Examples of the annotations of a meme and a video from the training sets are given in Figures 1 and 2.

```
"110001": {
  "id_EXIST": "110001",
  "lang": "es",
  "text": "2+2=5 MITO Albert Einstein tenia bajo rendimiento en la escuela. VERDAD 2+2=4 CAN is El feminismo de hoy en día defiende la estupidez humana y no los derechos de las mujeres quemó ellas afirman ",
  "meme": "110001.jpeg",
  "path_memes": "memes/110001.jpeg",
  "number_annotators": 6,
  "annotators": ["Annotator_1", "Annotator_2", "Annotator_3", "Annotator_4", "Annotator_5", "Annotator_6"],
  "gender_annotators": ["F", "F", "F", "M", "M", "M"],
  "age_annotators": ["18-22", "23-45", "46+", "46+", "18-22", "23-45"],
  "ethnicities_annotators": ["Hispano or Latino", "Hispano or Latino", "Hispano or Latino", "White or Caucasian", "Hispano or Latino", "Hispano or Latino"],
  "study_levels_annotators": ["High school degree or equivalent", "Master's degree", "Master's degree", "Bachelor's degree", "Bachelor's degree", "Bachelor's degree"],
  "countries_annotators": ["Mexico", "Spain", "Argentina", "Spain", "Mexico", "Mexico"],
  "labels_task2_1": ["YES", "YES", "YES", "YES", "YES", "YES"],
  "labels_task2_2": ["DIRECT", "DIRECT", "DIRECT", "DIRECT", "DIRECT", "DIRECT"],
  "labels_task2_3": [
    ["IDEOLOGICAL-INEQUALITY", "STEREOTYPING-DOMINANCE", "MISOGYNY-NON-SEXUAL-VIOLENCE"],
    ["IDEOLOGICAL-INEQUALITY"],
    ["IDEOLOGICAL-INEQUALITY"],
    ["IDEOLOGICAL-INEQUALITY"],
    ["IDEOLOGICAL-INEQUALITY"],
    ["IDEOLOGICAL-INEQUALITY"]
  ],
  "split": "TRAIN-MEME_ES"
},
```

Figure 1 Example of the annotations for a meme

```
"120001": {
  "id_Tiktok": 7281385962049998086,
  "id_EXIST": "120001",
  "lang": "es",
  "text": " cuando ves a las del 08 en la fiesta tu amigo el menos salido j5 [aza red. hay zorro al raimon ha conseguido impedir el tiro de la super táctica.",
  "video": "7281385962049998086.mp4",
  "path_video": "videos/7281385962049998086.mp4",
  "url": "https://www.tiktok.com/@inazumamemes/video/7281385962049998086",
  "annotators": ["Annotator_4", "Annotator_8", "Annotator_9"],
  "number_annotators": 3,
  "gender_annotators": ["F", "M", "M"],
  "labels_task3_1": ["YES", "NO", "YES"],
  "labels_task3_2": ["DIRECT", "-", "DIRECT"],
  "labels_task3_3": [
    ["OBJECTIFICATION"],
    ["-"],
    ["OBJECTIFICATION"]
  ],
  "split": "TRAIN-VIDEO_ES"
},
```

Figure 2 Example of the annotations for a video

Sensor-Based Experiments and Data Overview

This section presents the physiological (“sensorial”) information collected for the multimedia stimuli used in EXIST 2026, including image memes and short video items. These features are intended to complement the stimulus content with measurable signals of attention, arousal, and cognitive processing elicited while subjects viewed each stimulus.

During the experiment¹, subjects were seated comfortably while stimuli were displayed on a screen until they provided a response. A 3-second pause followed each stimulus to minimize carry-over effects between consecutive items. Each session lasted approximately 45 minutes, during which subjects viewed 100–170 stimuli. After each stimulus, they answered brief control questions to ensure engagement and comprehension. All participants provided consent for the anonymous use of their data for research purposes.

Where to Find the Sensorial Data in the JSON

For each instance in the dataset, the physiological data are stored under:

- instance["sensorial"]

And, inside this block, features are organized by subject and modality:

- sensorial["users"]
- sensorial["modalities"]["ET"]["by_user"]
- sensorial["modalities"]["HR"]["by_user"]
- sensorial["modalities"]["EEG"]["by_user"]

```
"countries_annotators": ["Spain", "Portugal", "Spain", "Canada", "United States", "Chile"],
"sensorial": {
  "users": ["ES1", "ES2", "ES3"],
  "modalities": {
    "ET": {
      "by_user": {
        "ES1": {
          "reaction_time": 23014.0,
          "3d_eye_states_pupil_diameter_left [mm]_mean": 2.7497,
          "3d_eye_states_pupil_diameter_left [mm]_std": 0.2256,
          "3d_eye_states_pupil_diameter_left [mm]_max": 4.2079,
          "3d_eye_states_pupil_diameter_left [mm]_min": 1.7904,
          "3d_eye_states_pupil_diameter_right [mm]_mean": 2.5381,
          "3d_eye_states_pupil_diameter_right [mm]_std": 0.1771,
          "3d_eye_states_pupil_diameter_right [mm]_max": 2.9745,
          "3d_eye_states_pupil_diameter_right [mm]_min": 1.2602,
          "blinks_duration_mean_ms": 226501504.0
        }
      }
    }
  }
}
```

Subjects per Stimulus

Each stimulus in the physiological study was viewed by multiple subjects. The **sensorial.users** list

¹ Data collection was approved by the Research Ethics Committee of the Universitat Politècnica de València (Project IDs P01_31-10-2025 and P13_26-12-2025).

contains the identifiers of all subjects associated with a given stimulus across the recorded modalities. In the released dataset, each stimulus has been viewed by **2–4 subjects**.

For each physiological modality—eye tracking (ET), heart rate (HR), and EEG—the corresponding **sensorial.modalities.<MOD>.by_user** dictionary indicates the subset of subjects for which that particular modality was recorded for the stimulus. As a consistency measure, the released files ensure that **each modality includes recordings from at least two subjects** for any stimulus where that modality is available. This guarantees that every stimulus has a minimum amount of data per modality while reflecting the actual number of subjects who participated in each recording.

What is Provided

For each subject and modality, we provide a compact dictionary of summary variables extracted from the physiological signals recorded while the subject viewed the stimulus. Concretely, for a subject identifier like ES1, the feature dictionary is:

- **sensorial.modalities.<MOD>.by_user["ES1"]**.

These variables are designed to be lightweight to use: they are aggregated descriptors of the stimulus viewing interval, defined as the time from stimulus onset until the subject advances to the next item by providing their response. The reported values correspond to aggregated statistics over this time window (e.g., mean, standard deviation, minimum, maximum, counts, or durations), rather than instantaneous measurements, and can be fed directly into standard ML pipelines as numerical features.

Modalities and Variables

Eye-Tracking (ET)

ET features summarize how visual attention unfolds while the subject views the stimulus. They can help model which content attracts attention, how strongly it captures gaze, and whether it increases visual/cognitive effort.

- Number of ET variables: 24

Variable groups:

- Pupil diameter (left/right): mean, std, min, max
- Blinks: count and duration statistics (mean, std, min, max)
- Fixations: count and duration statistics (mean, std, min, max)
- Saccades: count and duration statistics (mean, std, min, max)
- Reaction time: time to respond after viewing the stimulus

Why ET can help for sexism-related content: attention and viewing dynamics often change when stimuli contain offensive or emotionally salient cues. Pupil measures are widely used as proxies for cognitive effort and affective load; fixation and saccade patterns reflect how information is sampled; blink behavior can co-vary with attentional demand. Reaction time can reflect processing difficulty or hesitation after exposure to charged content.

```
"modalities": {
  "ET": {
    "by_user": {
      "ES1": {
        "reaction_time": 23014.0,
        "3d_eye_states_pupil diameter left [mm]_mean": 2.7497,
        "3d_eye_states_pupil diameter left [mm]_std": 0.2256,
        "3d_eye_states_pupil diameter left [mm]_max": 4.2079,
        "3d_eye_states_pupil diameter left [mm]_min": 1.7904,
        "3d_eye_states_pupil diameter right [mm]_mean": 2.5381,
        "3d_eye_states_pupil diameter right [mm]_std": 0.1771,
        "3d_eye_states_pupil diameter right [mm]_max": 2.9745,
        "3d_eye_states_pupil diameter right [mm]_min": 1.2602,
        "blinks_duration_mean_ns": 226501504.0,
        "blinks_duration_std_ns": 75922576.9786,
        "blinks_duration_max_ns": 350375168.0,
        "blinks_duration_min_ns": 145250048.0,
        "blinks_count": 4.0,
        "fixations_duration_mean_ns": 248963299.1549,
        "fixations_duration_std_ns": 220663904.2358,
        "fixations_duration_max_ns": 1351249920.0,
        "fixations_duration_min_ns": 69999104.0,
        "fixations_count": 71.0,
        "saccades_duration_mean_ns": 77578280.2286,
        "saccades_duration_std_ns": 106090482.0967,
        "saccades_duration_max_ns": 555500032.0,
        "saccades_duration_min_ns": 9999872.0,
        "saccades_count": 70.0
      },
      "ES2": {
```

Heart Rate (HR)

HR features summarize autonomic activation during stimulus viewing. They can help capture physiological arousal that may correlate with the emotional impact of the stimulus.

- Number of HR variables: 4

Variables:

- garmin_hr_mean
- garmin_hr_std
- garmin_hr_min
- garmin_hr_max

Why HR can help for sexism-related content: emotionally intense or disturbing material may increase arousal, which can appear as changes in heart-rate level and variability. These descriptors provide a compact signal of autonomic response that can complement purely content-based models.

Electroencephalography (EEG)

EEG features summarize brain activity during stimulus viewing. They can help capture differences in cognitive and affective processing that are not directly observable from the text or visuals alone.

- Number of EEG variables: 80

EEG variables are bandpower features computed for 16 channels (EXG_Channel_0 ... EXG_Channel_15), with 5 frequency bands per channel: Delta, Theta, Alpha, Beta, and Gamma bandpower.

Why EEG can help for sexism-related content: different types of stimuli can elicit distinct patterns of engagement, emotional reactivity, and cognitive control. Frequency-domain bandpower features provide compact descriptors of these processes and can support multimodal models that aim to detect or characterize sexist content beyond what is explicitly stated in the stimulus.

Questions

If you have any questions or problems, please open a thread on the support-forum on Discord.

References

Overview of Previous LeWiDi EXIST Editions:

- Laura Plaza, Jorge Carrillo-De-Albornoz, Iván Arcos, Paolo Rosso, Damiano Spina, Enrique Amigó, Julio Gonzalo, Roser Morante. [Overview of EXIST 2025: Learning with Disagreement for Sexism Identification and Characterization in Tweets, Memes, and TikTok Video](#). Carrillo-De-Albornoz, J., et al. Experimental IR Meets Multilinguality, Multimodality, and Interaction. CLEF 2025. Lecture Notes in Computer Science, vol 16089. Springer, Cham. <https://link.springer.com/book/10.1007/978-3-032-04354-2>.
- Laura Plaza, Jorge Carrillo-de-Albornoz, Victor Ruiz, Alba Maeso, Berta Chulvi, Paolo Rosso, Enrique Amigó, Julio Gonzalo, Roser Morante and Damiano Spina. (2024). [Overview of EXIST 2024 - Learning with Disagreement for Sexism Identification and Characterization in Tweets and Memes](#). In L. Goeuriot et al. (Eds.), Experimental IR Meets Multilinguality, Multimodality, and Interaction (pp. 93–117). Cham: Springer Nature Switzerland.

- Laura Plaza, Jorge Carrillo-de-Albornoz, Roser Morante, Enrique Amigó, Julio Gonzalo, Damiano Spina & Paolo Rosso . [Overview of EXIST 2023 – Learning with Disagreement for Sexism Identification and Characterization](#). Arampatzis, A., et al. Experimental IR Meets Multilinguality, Multimodality, and Interaction. CLEF 2023. Lecture Notes in Computer Science, vol 14163. Springer, Cham. https://doi.org/10.1007/978-3-031-42448-9_23.

Extended Overview of Previous LeWiDi EXIST Editions:

- Laura Plaza, Jorge Carrillo-De-Albornoz, Iván Arcos, Paolo Rosso, Damiano Spina, Enrique Amigó, Julio Gonzalo, Roser Morante. [Overview of EXIST 2025: Learning with Disagreement for Sexism Identification and Characterization in Tweets, Memes, and TikTok Videos \(Extended Overview\)](#). Guglielmo Faggioli et al. Working Notes of the Conference and Labs of the Evaluation Forum (CLEF 2025). CEUR Working Notes Vol-4038.
- Laura Plaza, Jorge Carrillo de Albornoz, Víctor Ruiz, Alba Maeso, Berta Chulvi, Paolo Rosso, Enrique Amigó, Julio Gonzalo, Roser Morante, Damiano Spina: [Overview of EXIST 2024 - Learning with Disagreement for Sexism Identification and Characterization in Tweets and Memes \(Extended Overview\)](#). Working Notes of the Conference and Labs of the Evaluation Forum (CLEF 2024). CEUR Working Notes Vol-3740.: 908-941
- Laura Plaza, Jorge Carrillo-de-Albornoz, Roser Morante, Enrique Amigó, Julio Gonzalo, Damiano Spina & Paolo Rosso . [Overview of EXIST 2023 – Learning with Disagreement for Sexism Identification and Characterization \(Extended Overview\)](#). Mohammad Aliannejadi, et al. Working Notes of the Conference and Labs of the Evaluation Forum (CLEF 2023). CEUR Working Notes Vol-3497.

Working Notes of Previous LeWiDi EXIST Editions:

- All 2025 proceedings available at: [EXIST 2025 Proceedings](#).
- All 2024 proceedings available at: [EXIST 2024 Proceedings](#).
- All 2023 proceedings available at: [EXIST 2023 Proceedings](#).

Video and Meme Related Work

- Iván Arcós and Paolo Rosso (2024) [Sexism Identification on TikTok: A Multimodal AI Approach with Text, Audio, and Video](#). In: Proc. 15th Int. Conf. of the CLEF Association, CLEF 2024, Grenoble, France, September 9-12, Springer-Verlag, LNCS(14958), pp. 61-73
- Italiani, P., Gimeno-Gómez, D., Ragazzi, L., Moro, G., Rosso, P. (2026). [MEMEWEAVER: Inter-Meme Graph Reasoning for Sexism and Misogyny Detection](#). In Findings EACL 2026.

Sensor Data and NLP Related Work

- Alaçam, Ö., Hoeken, S., & Zarrieß, S. (2024). [Eyes Don't Lie: Subjective Hate Annotation and Detection with Gaze](#). In Proceedings of EMNLP 2024 (pp. 187–205).
- Zhang, Y., Li, Q., Nahata, S., Jamal, T., Cheng, S., Cauwenberghs, G., & Jung, T.-P. (2024). [Integrating LLMs, EEG, and eye-tracking for word-level neural state classification](#). IEEE Transactions on Neural Systems and Rehabilitation Engineering, 32, 3465–3475.
- Oğuz, F. E., Alkan, A., & Schöler, T. (2023). [Emotion detection from ECG signals with different learning algorithms and automated feature engineering](#). Signal, Image and Video Processing, 17(7), 3783–3791.
- Yoon, H., Tolera, B. A., Gong, T., Lee, K., & Lee, S.-J. (2024). [By My Eyes: Grounding multimodal large language models with sensor data via visual prompting](#). In Proceedings of EMNLP 2024 (pp. 2219–2241).
- Kaixin Ji, Danula Hettiachchi, Falk Scholer, Flora D. Salim, and Damiano Spina. 2025. SenseSeek [Dataset: Multimodal Sensing to Study Information Seeking Behaviors](#). Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. 9, 3, Article 92 (September 2025), 29 pages. <https://doi.org/10.1145/3749501>